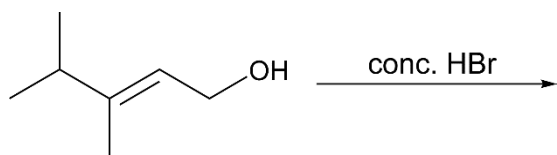
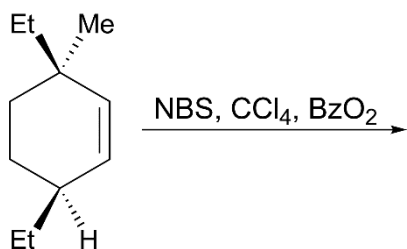


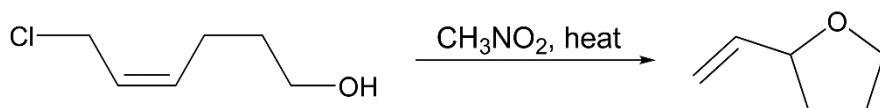
1. Complete the following and write a mechanism. Pay attention to the regiochemical outcome!



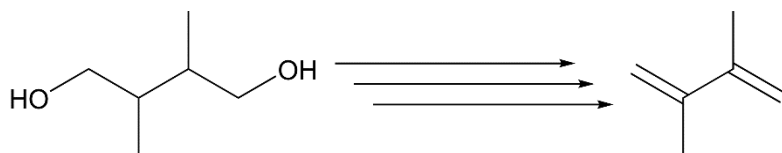
2. Complete the following and write a mechanism. Pay attention to the stereochemical outcome!



3. Draw a mechanism for the following transformation.



4. Propose reagents that could be used to complete the following synthesis.



5. Propose reagents that could be used to complete the following synthesis.

